Project Title

Data Analysis Project (Super Store)

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1: Project Overview:

This project analyzes sales and profitability performance using the Superstore dataset. The objective is to identify key business trends, evaluate customer and product performance, compare regional and category performance, and identify opportunities for improving profitability. SQL was used to perform data analysis and extract business insights, while Power BI was used to create an interactive dashboard for visualizing the results.

2: Dataset Description:

The Superstore dataset contains transactional sales data including orders, customers, products, categories, regions, sales, discounts, quantities, and profits. The dataset contains information about customer and product performance across different locations and time periods.

Main columns used in the analysis:

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Segment
- Region
- State
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

3: Tools & Technologies:

The project was completed using Microsoft Excel, python, SQL and Microsoft Power BI. Python and Excel were used during the initial data inspection and preparation stages, while SQL was used for the main data analysis and business queries. Power BI was then used to create the interactive dashboard and visualize the key findings. SQL was used to clean, analyze, aggregate, and explore the dataset, while Power BI was used to create interactive visualizations, KPI cards, slicers, and the final dashboard.

Tools used:

- MySQL / MySQL Workbench
- Python using Visual Studio
- Microsoft Excel
- Microsoft Power BI
- SQL
- DAX

4: Data Preparation & Cleaning:

Before performing the analysis, the dataset was reviewed to ensure that the data was suitable for analysis. The columns and data types were examined, missing values were checked, duplicate records and repeated order IDs were investigated, and numerical fields such as Sales and Profit were reviewed for unusual values. Negative profit values were retained because they represent actual loss-making transactions and are important for business analysis.

Important cleaning checks:

- Checked for missing values.
- Reviewed column data types.
- Examined duplicate and repeated Order IDs.
- Checked Sales and Profit values.
- Identified negative Profit values.
- Verified the dataset structure before analysis.

Important note:

Negative Profit was not removed.

This is important because removing negative profits would hide products, customers, or transactions that are generating losses.

5: SQL Analysis:

SQL was used to answer different business questions and calculate important performance metrics.

Main SQL analysis completed:

1. Total Sales
2. Total Profit
3. Total Orders
4. Total Customers
5. Average Order Value
6. Profit Margin
7. Sales by Category
8. Profit by Category
9. Sales by Region
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21. Monthly Sales
22. Monthly Profit
23. Running Sales
24. Running Profit
25. Customer First and Last Order Dates
26. Customer Active Period
27. Number of Orders per Customer

6: SQL Skills Learned:

During the project, I developed practical SQL skills for business data analysis and learned how to combine multiple SQL concepts to answer analytical questions.

Skills practiced:

- SELECT
- WHERE
- GROUP BY
- HAVING
- ORDER BY
- LIMIT
- DISTINCT
- COUNT()
- COUNT(DISTINCT)
- SUM()
- AVG()
- MIN()
- MAX()
- ROUND()
- CASE
- DATE_FORMAT()
- YEAR()
- DATEDIFF()
- JOIN
- INNER JOIN
- LEFT JOIN
- Common Table Expressions WITH
- Window functions
- LAG()
- SUM() OVER()
- PARTITION BY

- ORDER BY inside window functions

Important concepts learned:

GROUP BY was used to aggregate data by customers, products, categories, regions, segments, and time periods.

HAVING was used to filter aggregated results.

JOINs were used to combine related tables through common identifiers such as Customer_ID.

CASE was used to create business categories such as High Profit, Medium Profit, Loss, and Customer Value.

Window functions were used to calculate previous-period values and running totals without collapsing the original grouped results.

7: Key Business Metrics:

The analysis produced several important overall metrics.

Metric	Result
Total Sales	2,297,200.86
Total Profit	286,397.02
Total Orders	5,009
Total Customers	793
Average Order Value	458.61
Profit Margin	12.47%

These metrics provide a high-level overview of the company's overall sales and profitability performance.

8: Power BI Dashboard:

Power BI was used to transform the SQL analysis into an interactive business dashboard. The dashboard was designed to provide a clear overview of sales, profit, customers, products, categories, regions, segments, and trends.

Main dashboard components:

KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Profit Margin

Category Analysis

- Sales & Profit by Category
- Profit Margin by Category

Regional Analysis

- Sales by Region
- Profit Margin by Region

Segment Analysis

- Sales by Segment
- Profit by Segment

Customer Analysis

- Top 10 Customers by Sales
- Customer Sales vs Profit

Product Analysis

- Top 10 Products by Sales
- Top 10 Products by Profit
- Loss-Making Products

Trend Analysis

- Sales & Profit Trend
- Monthly Sales & Profit Trend

Interactive Filters

- Year
- Region
- Category

9: Key Business Insights:

1. Overall Business Performance

The Superstore dataset shows an overall Total Sales of **$2,297,200.86** and Total Profit of **$286,397.02**, resulting in an overall Profit Margin of **12.47%**.

The business generated a positive overall profit, but the relatively moderate profit margin shows that increasing sales alone is not enough to maximize business performance.

Profitability should be monitored alongside revenue to understand whether sales growth is generating healthy returns.

2. Category Performance

The three main categories generated the following sales:

Category	Total Sales
Technology	$836,154.03
Furniture	$741,999.80
Office Supplies	$719,047.03

Technology generated the highest sales, with approximately $836K, making it the largest category by revenue.

Furniture generated approximately $742K, while Office Supplies generated approximately $719K.

However, the analysis showed that sales ranking and profit ranking do not necessarily follow the same pattern. This is an important business insight because a category can generate substantial revenue without producing the same level of profitability.

Therefore, management should evaluate Profit and Profit Margin together with Sales when assessing category performance.

3. Regional Performance

The regional analysis shows clear differences in both sales and profitability across the four regions.

Region	Total Sales	Profit Margin
West	$725,457.82	14.94%
East	$678,781.24	13.48%
Central	$501,239.89	7.92%
South	$391,721.91	11.93%

Analysis

The West region generated the highest sales at $725,457.82 and also achieved the highest profit margin at 14.94%, making it the strongest-performing region overall. The East region ranked second in sales with $678,781.24 and achieved a profit margin of 13.48%.

The Central region generated $501,239.89 in sales but had the lowest profit margin at 7.92%. This indicates that despite generating more sales than the South region, its sales were converted into profit less efficiently.

The South region recorded the lowest sales at $391,721.91, but its profit margin of 11.93% was higher than Central's 7.92%. This demonstrates again that sales volume alone does not determine profitability.

4. Segment Performance

The segment analysis shows that the Consumer segment is the largest contributor to both sales and profit.

Segment	Total Sales	Sales %	Total Profit	Profit %
Consumer	$1,161,401.35	50.56%	$134,119.21	46.83%
Corporate	$706,146.37	30.74%	$91,979.13	32.12%
Home Office	$429,653.15	18.70%	$60,298.68	21.05%

Analysis

The Consumer segment generated the highest sales at $1,161,401.35, representing 50.56% of total sales. It also generated the highest total profit at $134,119.21, accounting for 46.83% of total profit. This makes Consumer the largest contributor to both revenue and profit.

The Corporate segment generated $706,146.37 in sales, representing 30.74% of total sales, and generated $91,979.13 in profit, representing 32.12% of total profit. Interestingly, its share of profit is higher than its share of sales, indicating relatively stronger profitability compared with its revenue contribution.

Home Office generated the lowest sales at $429,653.15, representing 18.70% of total sales, and generated $60,298.68 in profit, representing 21.05% of total profit. Similar to Corporate, its share of profit is higher than its share of sales, suggesting that the segment contributes to profitability at a higher proportion than its sales contribution.

5. Customer Performance

The customer analysis showed significant differences between customers in terms of sales and profitability.

For example, some customers generated particularly high profits:

Customer	Total Sales	Total Profit	Profit Margin
Tamara Chand	$19,052.22	$8,981.32	47.14%
Raymond Buch	$15,117.34	$6,976.10	46.15%

These examples demonstrate that some customers generate both strong sales and strong profitability.

We also analyzed customers with high sales but low profit, which is important because high-value customers based on revenue alone may not necessarily be the most valuable from a profitability perspective.

The Customer Sales vs Profit scatter chart was therefore used to identify the relationship between customer revenue and profitability.

6. Product Performance

The product analysis was divided into three perspectives: top products by sales, top products by profit, and loss-making products. This allows the business to distinguish between products that generate high revenue and products that generate high profitability.

Top Products by Sales

Product	Total Sales
Canon imageCLASS 2200 Advanced Copier	$61,599.82
Fellowes PB500 Electric Punch Plastic Comb	$27,453.38
Cisco TelePresence System EX90 Videoconferencing Unit	$22,638.48

The Canon imageCLASS 2200 Advanced Copier generated the highest sales among the products shown, with $61,599.82 in total sales. The Fellowes PB500 Electric Punch Plastic Comb ranked second with $27,453.38, while the Cisco TelePresence System EX90 Videoconferencing Unit generated $22,638.48. The large difference between the first product and the other products indicates that a small number of products can contribute substantially to total revenue.

Top Products by Profit

Product	Total Profit
Canon imageCLASS 2200 Advanced Copier	$25,199.93
Fellowes PB500 Electric Punch Plastic Comb	$7,753.04
Hewlett Packard LaserJet 3310 Copier	$6,983.88

The Canon imageCLASS 2200 Advanced Copier also generated the highest profit among the products shown, with $25,199.93. The Fellowes PB500 Electric Punch Plastic Comb generated $7,753.04 in profit, followed by the Hewlett Packard LaserJet 3310 Copier with $6,983.88. The Canon copier therefore stands out as a strong product in both revenue generation and profitability.

The product analysis demonstrates that the products generating the highest sales are not always the same products generating the highest profit. However, the Canon imageCLASS 2200 Advanced Copier is a strong exception in this analysis because it appears among the leading products for both sales and profit. At the same time, the loss-making products highlight the importance of monitoring profitability at the individual product level rather than focusing only on revenue.

7. Loss-Making Products

Product	Total Profit
Cubify CubeX 3D Printer Double Head Print	-$8,879.97
Lexmark MX611dhe Monochrome Laser Printer	-$4,589.97
Cubify CubeX 3D Printer Triple Head Print	-$3,839.99

The loss-making products analysis identified several products with negative total profit. The Cubify CubeX 3D Printer Double Head Print recorded the largest loss at -$8,879.97, followed by the Lexmark MX611dhe Monochrome Laser Printer at -$4,589.97 and the Cubify CubeX 3D Printer Triple Head Print at -$3,839.99. These products should be investigated further to determine the factors contributing to their negative profitability, such as discounts, pricing, or product-related costs.

8. Yearly Sales Performance

The yearly analysis showed changes in sales performance over time.

From the yearly analysis used in the project, sales increased substantially after the decline observed in 2015.

The calculated year-over-year changes included:

- 2015: Sales decreased by 2.83%
- 2016: Sales increased by 29.47%

This indicates that the business experienced a strong recovery and growth period after 2015.

The analysis also showed strong growth in annual profit, with profit increasing from:

- 2014: $46,970.95
- 2015: $50,466.07
- 2016: $60,362.31
- 2017: $128,597.70

The particularly strong profit result in 2017 indicates a major improvement in profitability during the final year of the dataset.

9. Monthly Sales Performance

The monthly analysis revealed clear differences in sales throughout the year.

Some of the strongest sales months were:

Month	Sales
September	$307,649.95
November	$352,461.07

December	$325,293.50

September, November, and December were among the strongest-performing months.

This suggests that the business experiences periods of higher demand and that seasonal patterns could be considered when planning inventory, marketing campaigns, and sales activities.

10. Sales vs. Profit

One of the main findings from the analysis is that high sales do not always translate into high profitability. The comparison between Sales and Profit across customers, products, categories, and regions shows that revenue alone is not sufficient to evaluate business performance. Therefore, Profit and Profit Margin should be considered alongside Sales when evaluating business performance and making decisions.

11. Overall Insight

The analysis shows that the Superstore business has strong overall revenue and positive profitability, but performance varies considerably across categories, customers, products, regions, segments, and time periods.

The most important business lesson from the project is that revenue should not be evaluated independently from profitability. Identifying highly profitable customers and products, understanding loss-making products, monitoring profit margins, and analyzing seasonal trends can help the business make more informed decisions and improve its overall profitability.

10: Yearly Performance:

Year	Sales
2014	484,247.50
2015	470,532.51
2016	609,205.60
2017	733,215.26

- 2015: decrease around 2.83%
- 2016: increase around 29.47%
- 2017: increase around 20.36%

The yearly analysis shows that sales declined slightly in 2015 before experiencing strong growth in 2016. This indicates a significant improvement in sales performance after 2015.

11: Monthly Performance:

The monthly analysis was used to identify periods of higher and lower sales and profit performance throughout the year. The monthly trend visualization makes it easier to identify seasonal patterns and changes in business performance over time.

- September
- November
- December

12: Profitability Analysis:

Profitability was analyzed using total profit and profit margin rather than relying only on sales. This helped identify categories, customers, regions, and products that generate strong revenue but may have relatively low profitability.

*High Sales does not always mean High Profit.

13: Business Recommendations:

1. Focus on profitable products

Identify and prioritize products that consistently generate high profit and healthy profit margins.

2. Investigate loss-making products

Review pricing, discounts, shipping costs, and other factors affecting products with negative total profit.

3. Evaluate high-sales but low-profit customers

Customers generating high sales but low profit should be investigated to understand whether discounts or other factors are reducing profitability.

4. Analyze regional performance

Regions with lower profit margins should be investigated to identify operational or pricing factors affecting profitability.

5. Use seasonal trends for planning

Monthly sales patterns can be used to improve inventory planning, marketing campaigns, and sales strategies during high-demand periods.

6. Monitor profit margin alongside sales

Management should monitor both revenue and profitability to avoid focusing on sales growth without considering its impact on profit.

14: Conclusion:

This project provided practical experience in using SQL and Power BI to perform end-to-end business data analysis. The analysis covered sales, profit, customers, products, categories, regions, segments, and time-based trends. SQL was used to extract and analyze the data, while Power BI was used to transform the findings into an interactive dashboard.

The project also demonstrated the importance of analyzing profitability alongside sales. High sales do not always result in high profit, making profit margin, customer profitability, product profitability, and loss analysis important for business decision-making**.**

15: Project Outcome:

The final result is a complete Data Analyst portfolio project combining SQL analysis and Power BI visualization. The project demonstrates practical skills in data preparation, SQL querying, aggregation, joins, window functions, business analysis, DAX, dashboard development, data visualization, and communicating business insights.